

Problem 1: Brill's Tagger

(15 marks)

A part-of-speech (POS) tagger is a piece of software that takes a sentence as input and outputs a tag for each word indicating the part of speech: broadly speaking, noun, verb, preposition, etc. One of the first statistical taggers ('statistical' meaning it was trained on sample data) was developed by Eric Brill in 1995 and is popularly known as Brill's Tagger. This worked in two steps. In the first step, each word was assigned the most likely tag, irrespective of context. For example, since 'book' as a noun is more common than 'book' as a verb, the first step would always mark the word 'book' in any sentence as a noun (symbol NN). Then, in the second step, some rules are applied to change the tags of some words based on context.

In the tables below, the first row has the original sentences, the second row has the result of the first step (assigning tags), and the third row has the result of the second step (applying rules). Note that some of the tags have been omitted. You may be able to figure out the meanings of some of the tags, but that is not required to solve this problem.

(a) Saujas books one ticket for the show .
 NNP NNS CD NN PP DT VF PU
 NNP **(1)** CD NN PP DT NN PU

(b) Since the show is on for two days ,
 PP DT VF VF PP **(2)** CD NNS PU
 CNJ DT NN VF PT PP CD NNS PU
 he could go on both days .
 PRP VM VF PP DT NNS PU
 PRP VM VF PP DT **(3)** PU

(c) Saujas watches it on the first day .
 NNP NNS PT PP DT OD NN PU
 NNP VF **(4)** PP DT OD NN PU

(d) He suggests Shardul to watch it as well .
 PRP **(5)** NNP PT VF PT RB **(6)** PU
 PRP VF NNP **(7)** VF **(8)** RB RB PU

(e) Shardul books another ticket and watches it the next day .
 NNP NNS DT NN CNJ NNS **(9)** DT JJ NN PU
 NNP VF DT NN CNJ RB **(10)** DT JJ NN PU

The following rules are applied one by one in the given order. Note that some parts of the rules have been omitted.

- (A) PT PRP WDPREVTAG VF it
- (B) PP **(11)** **(12)** DT since
- (C) PP PT WDPREVTAG VF WDNEXTTAG PP
- (D) NNS VF **(13)** NNP
- (E) VF NN WDPREVTAG **(14)**
- (F) NNS **(15)** WDNEXTTAG PT

Assignment 1: (12 marks)

Fill in the omitted entries numbered (1)–(15). It may be helpful to know that rule (C) affects only one word and rule (E) does not affect sentence (c).

Assignment 2: (3 marks)

What will Brill's Tagger output for this sentence in both steps?

"He books it for another show."

– *Monojit Choudhury*